onto the black hole, or if two black holes collide and combine—in any of these processes and all others as well, the total area of the event horizon of a black hole always increases. To Bekenstein, the inexorable evolution to greater total area suggested a link with the inexorable evolution to greater total entropy embodied in the second law of thermodynamics. He proposed that the area of the event horizon of a black hole provides a precise measure of its entropy.

On closer inspection, though, there are two reasons why most physicists thought that Bekenstein's idea could not be right. First, black holes would seem to be among the most ordered and organized objects in the whole universe. Once one measures the black hole's mass, the force charges it carries, and its spin, its identity has been nailed down precisely. With so few defining features, a black hole appears to lack sufficient structure to allow for disorder. Just as there is little havoc one can wreak on a desktop that holds solely a book and a pencil, black holes seem too simple to support disorder. The second reason that Bekenstein's proposal was hard to swallow is that entropy, as we have discussed it here, is a quantum mechanical concept, whereas black holes, until recently, were firmly entrenched in the antagonistic camp of classical general relativity. In the early 1970s, without a way to merge general relativity and quantum mechanics, it seemed awkward, at best, to discuss the possible entropy of a black hole.

How Black Is Black?

As it turns out, Hawking too had thought of the analogy between his black hole area-increase law and the law of inevitable increase of entropy, but he dismissed it as nothing more than a coincidence. After all, Hawking argued, based upon his area-increase law and other results he had found with James Bardeen and Brandon Carter, if one did take the analogy between the laws of black holes and the laws of thermodynamics seriously, not only would one be forced to identify the area of the black hole's event horizon with entropy, but it turns out that one would also have to assign a temperature to the black hole (with the precise value determined by the strength of the black hole's gravitational field at its event horizon). But if a black hole has a nonzero temperature—no matter how small—the most basic and well-established physical principles would require it to emit radiation, much